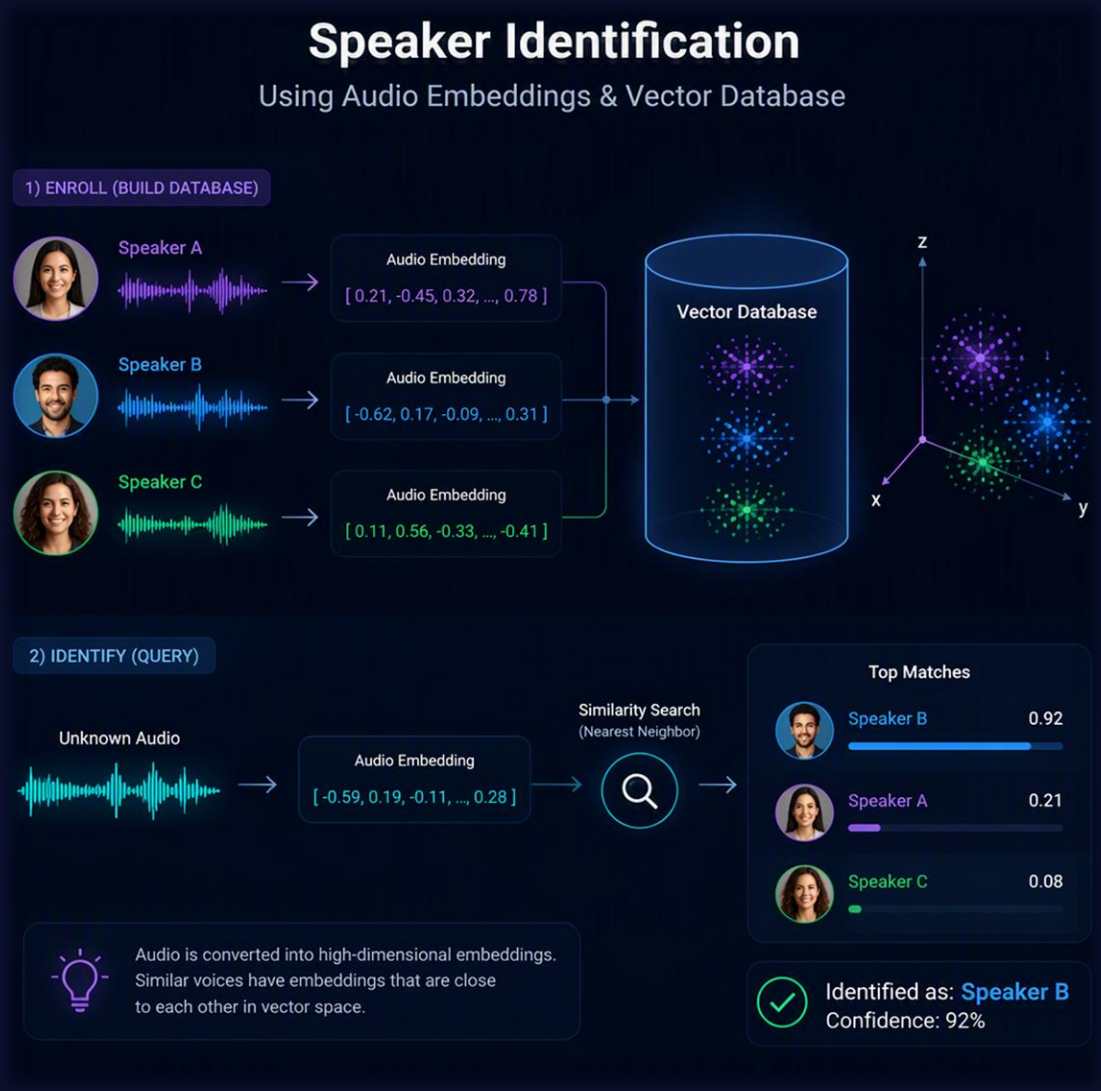
Microsoft

VOICE & AUDIO AI

Speaker Identification

with Audio Embeddings and Vector Search

A portable, open pipeline that recognizes who is speaking, built with audio embeddings and a vector database.



GitHub repo:
https://github.com/retkowsky/speaker-identification

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The problem & the gap

THE PROBLEM

Recognizing a known speaker from raw audio

Many applications need to answer one question from a voice recording: who is speaking? Customer service, meeting platforms, media archives, and security systems all depend on it.

1

A known set of speakers

Match voices against a controlled roster of enrolled individuals, not open-world transcription.

2

Deployment flexibility

Some projects run entirely in the cloud, other on-premises, at the edge, or fully offline.

3

Fast, trustworthy decisions

Identification must return a clear confidence score an application or operator can trust.

WHY IT MATTERS

Three building blocks combine into one pipeline: audio embeddings, vector search, edge deployment if required.

Two interchangeable vector back-ends: **Azure AI Search** for cloud scale, **Qdrant** for local and disconnected use.



THE GAP

About Speaker Recognition

Azure AI Speech keeps a broad voice portfolio, but the **managed speaker recognition API is being retired**.

BROADLY AVAILABLE IN AZURE AI SPEECH

- Speech to text (standard and custom)
- Text to speech (neural voices)
- Language identification and translation
- Speaker diarization (who spoke when)

Speaker identification is the one capability missing from this list.

RETIRING **Sept. 30, 2025**

Azure AI Speaker Recognition is retired, and applications can no longer access its APIs.

MICROSOFT GUIDANCE

Migrate to another solution available in the market, as early as possible, to allow time for transitioning enrolled profiles and for tuning applications to the new accuracy and performance characteristics.

Source: Azure updates, Microsoft Azure
Azure updates | Microsoft Azure

The solution

THE APPROACH

Build identification with embeddings and vector search

A fully portable, three-step pipeline built from open building blocks that run on any infrastructure and any deployment model.

01

Extract embeddings

Convert each enrolled voice sample into a fixed-length audio embedding vector.

SpeechBrain ECAPA-VOXCELEB model: https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb

02

Index in a vector store

Persist enrolled embeddings with their speaker label in **Azure AI Search or Qdrant**.

03

Match by similarity

Embed the unknown audio and run a nearest-neighbor search to return the closest speaker and a confidence score.

TWO INTERCHANGEABLE VECTOR BACK-ENDS

- **Azure AI Search** enterprise-scale, cloud-connected deployments
- **Qdrant** local, offline, disconnected or edge deployments

ARCHITECTURE

End-to-end architecture

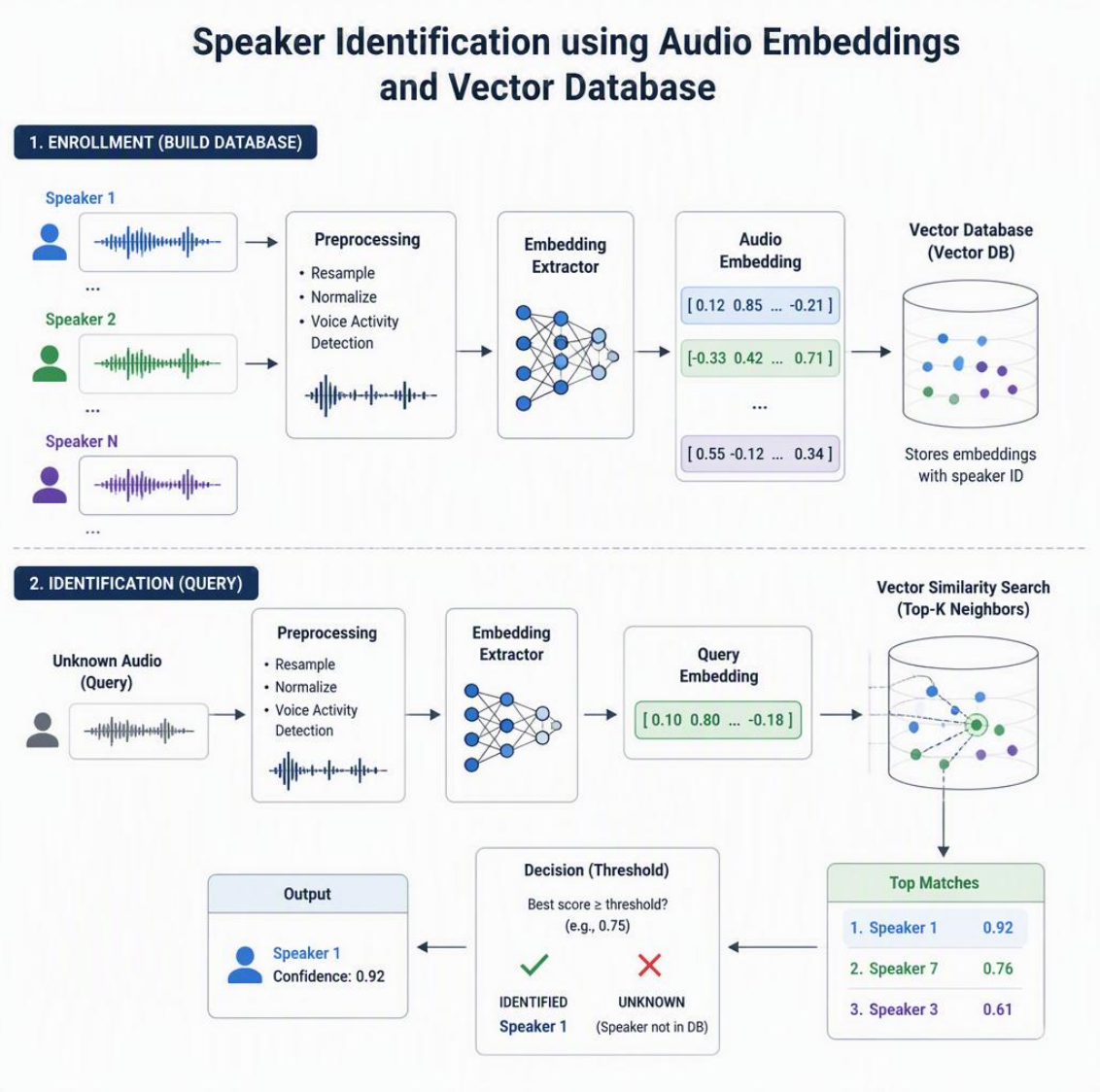
Enrollment builds the vector database; identification queries it in real time. Both phases share the same preprocessing and the same embedding extractor.

1 Enrollment

Voice samples are embedded once and stored with their speaker ID.

2 Identification

The query embedding is compared to the store with a top-K search.



RESULTS

Results and key takeaways

Two working end-to-end proofs of concept: one on Azure AI Search, one on a fully local Qdrant setup.

- Enrolled speakers are correctly identified from unseen audio, with a similarity score returned alongside each match.
- The same embeddings and matching logic run unchanged against either vector back-end, so the choice is a deployment decision, not a rewrite.
- Predictions are exported to spreadsheets (speakers_predictions.xlsx and speakers_predictions_qdrant.xlsx) for review and audit.

EXAMPLE MATCH

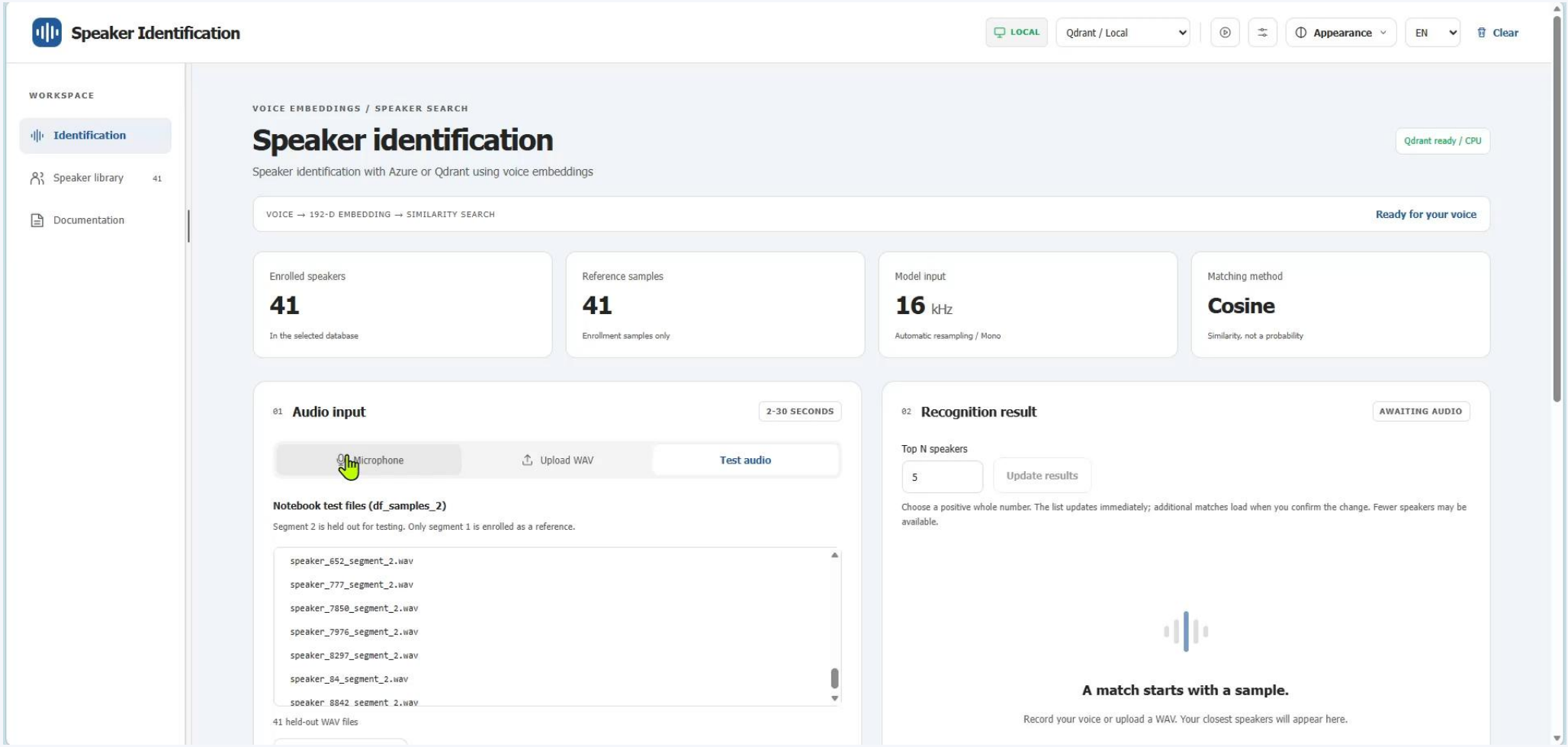
Voice is identified as Jane



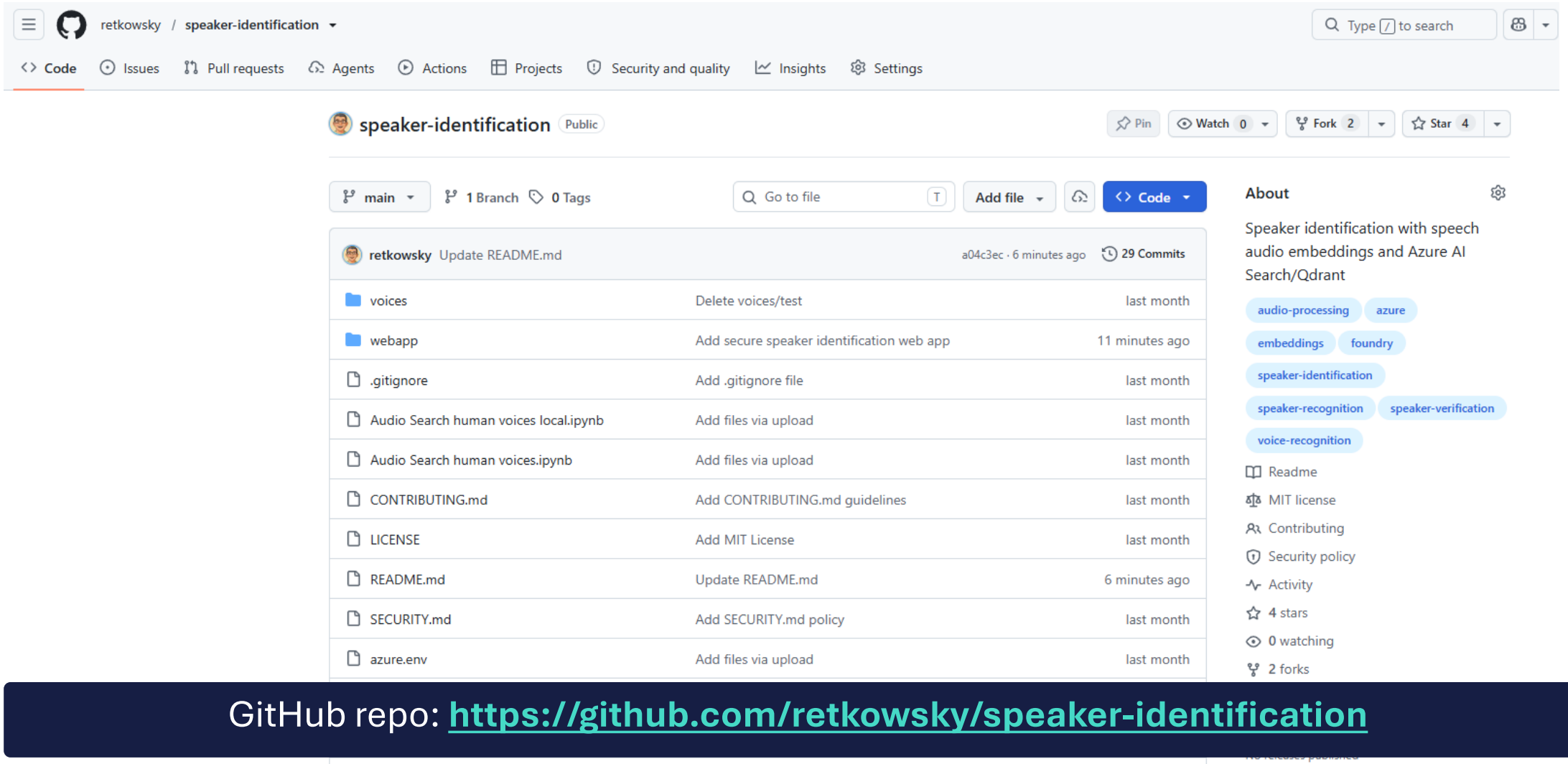
Cosine similarity over enrolled voice embeddings, threshold at 0.75.

RESULTS

The IP webapp




The IP



GitHub repo: https://github.com/retkowsky/speaker-identification

Thank you

